A brief Introduction to String theory

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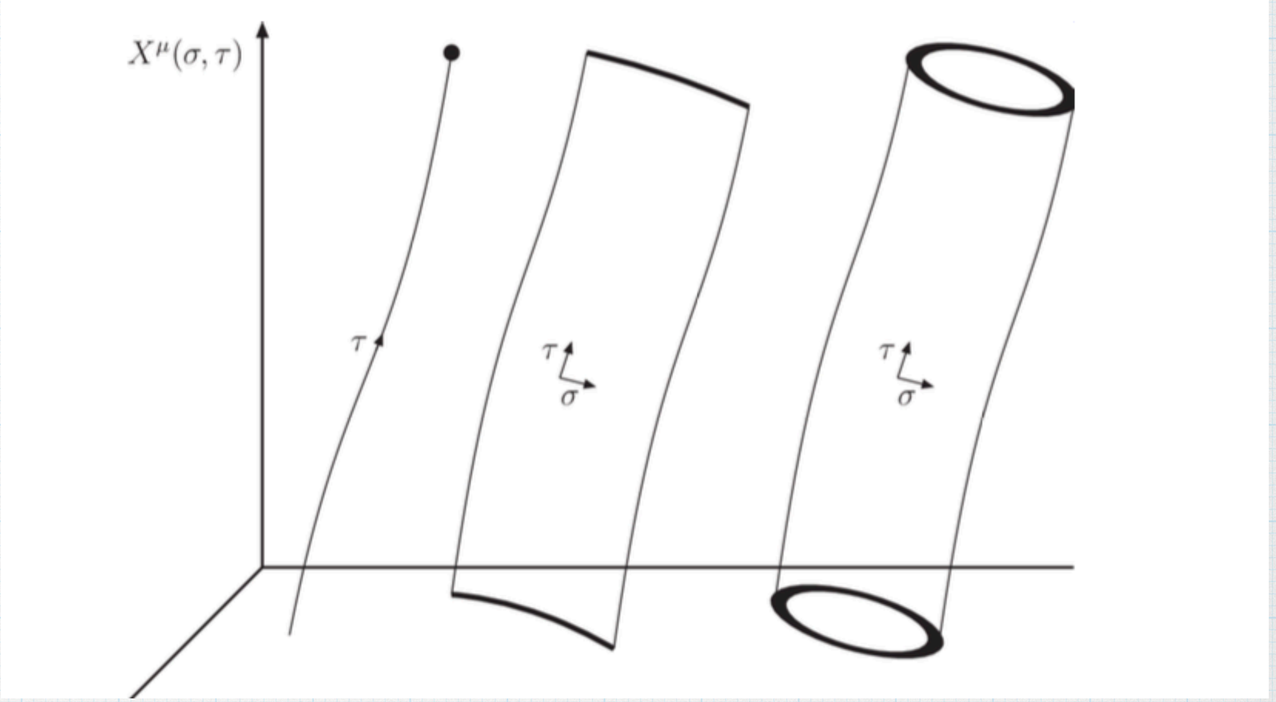
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Why String theory?

* Can we unify all fundamental interactions at quantum level?
* It has worked previously - electromagnetism, electro-weak interactions etc.
* Unify gravity at quantum level.
* Explaining problems of physics like dark matter, inflation, hierarchy of different scales etc.

Bosonic String theory

* Replace a particle with an extended object: String



$$S_{NG} = -T\int d\tau d\sigma\sqrt{-g} \quad \leftrightarrow \quad S_P = -\frac{T}{2}\int d\tau d\sigma\sqrt{-det(h)}h^{\alpha\beta}\,\partial_\alpha X^\mu\,\partial_\beta X_\nu\,\eta_{\mu\nu}$$

Nambu-Goto ↔ Polyakov

Properties of string action

* String world sheet is two dimensional.
* Poincare Invariance: $X_\mu \to \Lambda^\mu_\nu X^\nu + c_\mu \, .$
* Reparameterization Invariance: $\sigma_\alpha \to \tilde{\sigma}_\alpha(\sigma)$
* Weyl Invariance : $g_{\alpha\beta}(\sigma) \to \Omega^2(\sigma) g_{\alpha\beta}(\sigma)$

Extra Dimensions

* After quantisation, one needs to impose extra conditions to preserve Weyl invariance.

* The spectrum of string is determined by:

$$M^2 = \frac{4}{\alpha'}\left(N - \frac{D-2}{24}\right), \qquad N = \sum_{i=1}^{D-2}\sum_{n>0} \alpha^i_{-n}\alpha^i_n .$$

* In order to preserve, $SO(1,D-1)$ Lorentz symmetry, we need

$$D = 26!!$$

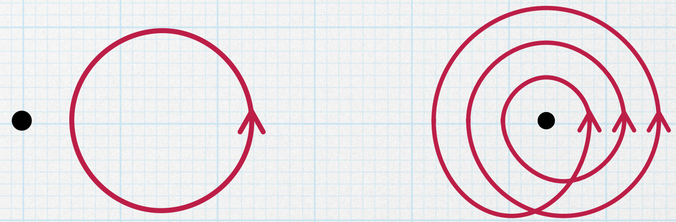
Superstring theory

* Superstring is the addition of fermionic modes on the worldsheet.
* The critical dimension of the superstring is $D = 10$..
* Type II strings have both left and right-moving worldsheet fermions; $D = 10\,, \mathcal{N} = 2\,.$.
* Heterotic strings have just right-moving fermions; $\mathcal{N} = 1\,.$

Dualities

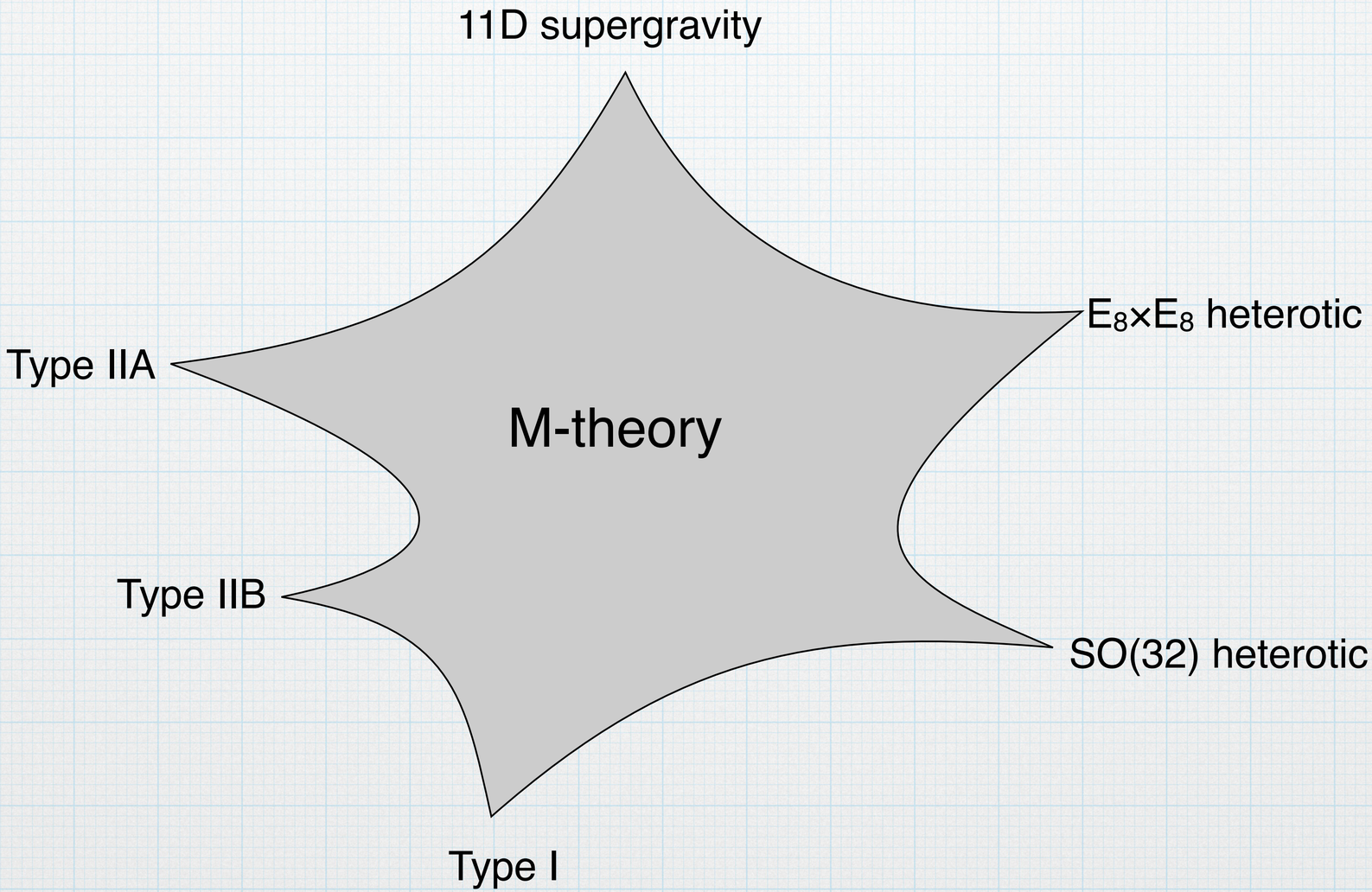
* **T-duality:** Take one of those nine space dimensions and make it a circle of radius R ⟶ Winding Number

$$(mR)^2 + \left(\frac{n}{R}\right)^2 , \qquad R \leftrightarrow \frac{1}{R} \;\&\; m \leftrightarrow n \,.$$



* **S-duality:** Large Coupling ⟷ Small Coupling or g ⟷ 1/g.

M-theory


11D supergravity
Type IIA
M-theory
$E_8 \times E_8$ heterotic
Type IIB
SO(32) heterotic
Type I

What do we want to achieve?

* String theory being a candidate for a unified theory must be able to reproduce the physics of our real world.
* The final solution should be four dimensional.
* Supersymmetry must be broken.
* Should come with dark matter, inflation etc.
* Vacuum should be de-Sitter.

Compactification

* Take a free scalar field in 5-dimensions $\mathscr{M} = \mathbb{R}^4 \times S^1$, the action is

$$S = \frac{1}{2}\int d^5x \partial_N \phi \partial^N \phi \qquad \partial^N \partial_N \phi = \partial^\mu \partial_\mu \phi + \partial^5 \partial_5 \phi = 0$$

* Since x5 $\in$ (0,2πR), expand the scalar field into Fourier series

$$\phi(\mathbf{x}) = \sum_{\mathbf{n}=\mathbf{0}}^{\infty} \phi_{\mathbf{n}} \mathbf{e}^{\mathbf{i n x^5/R}}, \qquad \partial^\mu \partial_\mu \phi_{\mathbf{n}} + \left(\frac{\mathbf{n}}{\mathbf{R}}\right)^{\mathbf{2}} \phi_{\mathbf{n}}$$

* Take $R \sim 10^{-33} cm$, focus on zero modes.

Compactification: Moduli Fields

* Take pure gravity in 5-d,

$$S = \frac{M_{P,5}^3}{2} \int d^4x dy \sqrt{-g_5} \mathscr{R}_5$$

* Ansatz for metric,

$$g_{MN} = \begin{pmatrix} e^{2\alpha\phi} g_{\mu\nu} + e^{2\beta\phi} \mathscr{A}_\mu \mathscr{A}_\nu & e^{2\beta\phi} \mathscr{A}_\mu \\ e^{2\beta\phi} \mathscr{A}_\nu & e^{2\beta\phi} \end{pmatrix},$$

* Focusing on zero modes gives massless scalars,

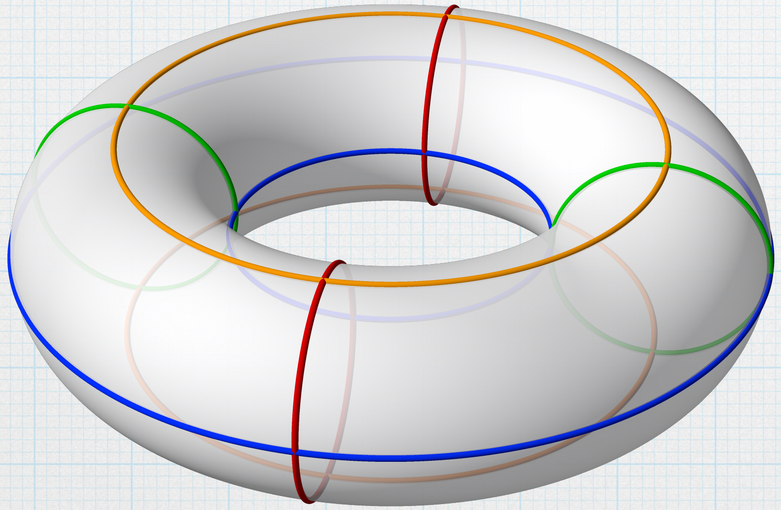
$$S = \frac{M_{P,5}^3}{2} \int d^4x \sqrt{-g} \left(\mathscr{R} - \frac{1}{2} (\partial\phi)^2 - \frac{1}{4} e^{-6\alpha\phi} \mathscr{F}^2 \right)$$

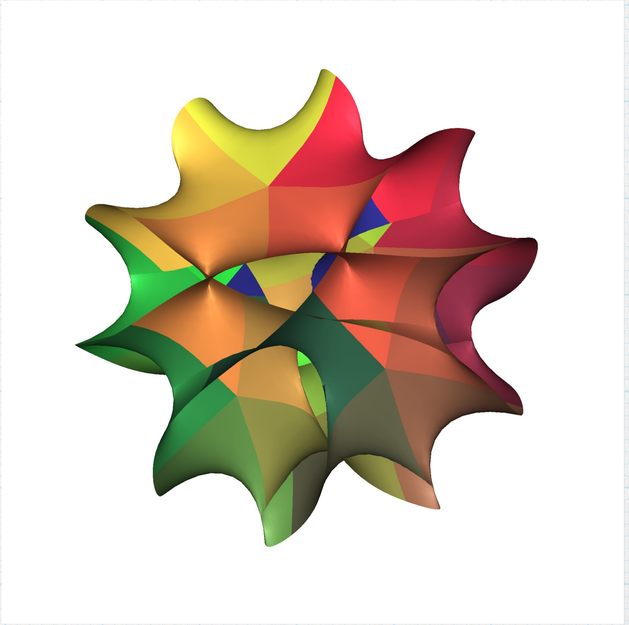
String Phenomenology

* We want to find a solution of 10d equation of motion corresponding to a 4d world.
* A three complex dimensional Calabi-Yau manifold is a well motivated candidate for compactification because it gives rise to MSSM in 4-dimensional.
* Calabi-Yau compactification comes with large number of moduli.

A bit more about Calabi-Yau

* Complex, Kahler, Ricci Flat manifold.
* Preserves SUSY.

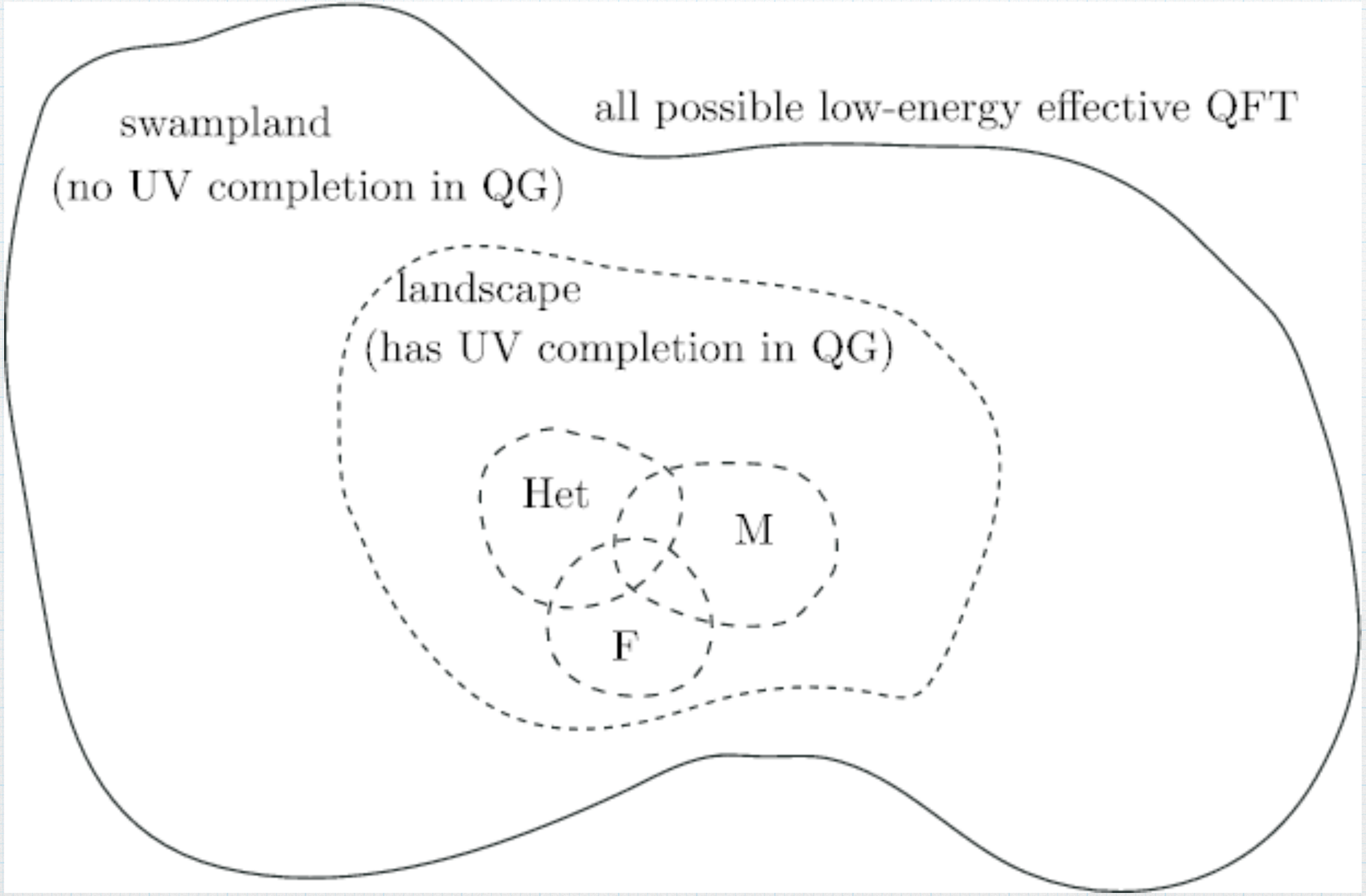




Moduli Stabilisation

* Moduli couple to ordinary matter via gravitation, so could give rise to forces (so far not detected).
* We need to stabilise moduli ⟶ Done by Introducing quantised fluxes.
* Very Large number of configurations, **String Landscape** $\sim 10^{272,000}$ **vacua**.

Remarks: On Going Research



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Landscape	Swampland
Can we have a metastable de Sitter?	Can all consistent low energy effective field theories that include gravity actually arise in string theory or not?
Non-SUSY Compactifiactions?	Conjectures and their implications for cosmology and particle physics?

Thank you.